

```
# This R environment comes with many helpful analytics packages installed
# It is defined by the kaggle/rstats Docker image: https://github.com/kaggle/docker-rstats
# For example, here's a helpful package to load

library(tidyverse) # metapackage of all tidyverse packages

# Input data files are available in the read-only "../input/" directory
# For example, running this (by clicking run or pressing Shift+Enter) will list all files under the input directory

list.files(path = "../input")

# You can write up to 20GB to the current directory (/kaggle/working/) that gets preserved as output when you create a version
# You can also write temporary files to /kaggle/temp/, but they won't be saved outside of the current session
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.1
✓ ggplot2     3.5.1    ✓ tibble     3.2.1
✓ lubridate   1.9.4    ✓ tidyr      1.3.1
✓ purrr       1.0.2
— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (http://conflicted.r-lib.org/) to force all conflicts to become errors
```

Lab Problem Statement 6: Statistical Analysis of Physical Characteristics of Palmer Penguins

Objective

To perform statistical analysis on the Palmer Penguins dataset to determine whether physical characteristics such as body mass and flipper length differ significantly across penguin species and sex.

Statistical Methods

- Descriptive Statistics
- Histogram, Boxplot and Density Plot
- Shapiro-Wilk Normality Test
- Q-Q Plot
- Independent Two-Sample Welch t-Test
- 95% Confidence Interval
- Cohen's d Effect Size
- One-Way ANOVA
- Levene's Test
- Tukey's HSD Post-Hoc Test
- Kruskal-Wallis Test
- Two-Way ANOVA
- Interaction Analysis
- Flipper-Length Analysis

Dataset

The Palmer Penguins dataset contains physical measurements of 344 penguins belonging to three species:

- Adelie
- Chinstrap
- Gentoo

Variables Used

Variable	Description
species	Penguin species
island	Island where the penguin was observed
sex	Penguin sex
bill_length_mm	Bill length in millimetres
bill_depth_mm	Bill depth in millimetres
flipper_length_mm	Flipper length in millimetres

Variable	Description
body_mass_g	Body mass in grams

1. Setup and Required Packages

```
# Install required packages
required_packages <- c(
  "palmerpenguins",
  "tidyverse",
  "moments",
  "car",
  "rstatix",
  "effectsize"
)

for (pkg in required_packages) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg, repos = "https://cloud.r-project.org")
  }
}

# Load packages
library(palmerpenguins)
library(tidyverse)
library(moments)
library(car)
library(rstatix)
library(effectsize)

# Reproducibility
set.seed(123)
```

2. Load the Palmer Penguins Dataset

```
# Load the Palmer Penguins dataset
data("penguins")

# Create working copy
penguin_data <- penguins

# Display first six observations
head(penguin_data)
```

A tibble: 6 × 8

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
<fct>	<fct>	<dbl>	<dbl>	<int>	<int>	<fct>	<int>
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18.0	195	3250	female	2007
Adelie	Torgersen	NA	NA	NA	NA	NA	2007
Adelie	Torgersen	36.7	19.3	193	3450	female	2007
Adelie	Torgersen	39.3	20.6	190	3650	male	2007

```
# Dataset dimensions
cat("Number of observations:", nrow(penguin_data), "\n")
cat("Number of variables:", ncol(penguin_data), "\n")
```

```
# Dataset structure
str(penguin_data)
```

```
Number of observations: 344
Number of variables: 8
tibble [344 × 8] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm: num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm: num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g   : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex          : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year         : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

3. Data Exploration and Missing Values

```
# Summary of dataset
summary(penguin_data)
```

```
# Missing values in each variable
colSums(is.na(penguin_data))
```

```

      species      island  bill_length_mm  bill_depth_mm
Adelie   :152  Biscoe   :168    Min.   :32.10    Min.   :13.10
Chinstrap: 68  Dream    :124    1st Qu.:39.23    1st Qu.:15.60
Gentoo   :124  Torgersen: 52    Median :44.45    Median :17.30
                                     Mean   :43.92    Mean   :17.15
                                     3rd Qu.:48.50    3rd Qu.:18.70
                                     Max.   :59.60    Max.   :21.50
                                     NA's   :2        NA's   :2

 flipper_length_mm  body_mass_g      sex      year
Min.   :172.0      Min.   :2700   female:165  Min.   :2007
1st Qu.:190.0      1st Qu.:3550   male :168   1st Qu.:2007
Median :197.0      Median :4050   NA's  : 11   Median :2008
Mean   :200.9      Mean   :4202                   Mean :2008
3rd Qu.:213.0      3rd Qu.:4750                   3rd Qu.:2009
Max.   :231.0      Max.   :6300                   Max.   :2009
NA's   :2          NA's   :2

species:      0 island:      0 bill_length_mm:      2 bill_depth_mm:      2 flipper_length_mm:      2 body_mass_g:      2 sex:      11 year:
```

```
# Species frequency
table(penguin_data$species)
```

```
# Sex frequency
table(penguin_data$sex)
```

```
# Species and sex frequency
table(penguin_data$species, penguin_data$sex)
```

```

      Adelie Chinstrap  Gentoo
      152      68      124

female  male
 165    168

      female male
Adelie      73  73
Chinstrap    34  34
Gentoo      58  61
```

4. Data Cleaning

Rows containing missing values in the variables required for the analysis are removed.

```
# Select variables required for analysis
analysis_data <- penguin_data %>%
  select(
    species,
    island,
    sex,
    bill_length_mm,
    bill_depth_mm,
    flipper_length_mm,
    body_mass_g
  ) %>%
  drop_na()
```

```
# Check cleaned dataset
dim(analysis_data)
```

```
# Check remaining missing values
colSums(is.na(analysis_data))
```

```

333 · 7
species:      0 island:      0 sex:      0 bill_length_mm:      0 bill_depth_mm:      0 flipper_length_mm:      0 body_mass_g:      0
```

Task 1: Descriptive Statistical Analysis

Descriptive statistics are calculated for penguin body mass.

The following measures are calculated:

- Mean
- Median
- Minimum
- Maximum
- Variance
- Standard Deviation
- First Quartile
- Third Quartile
- Interquartile Range
- Skewness
- Kurtosis

5. Overall Descriptive Statistics of Body Mass

```
body_mass_summary <- analysis_data %>%
  summarise(
    N = n(),
    Mean = mean(body_mass_g),
    Median = median(body_mass_g),
    Minimum = min(body_mass_g),
    Maximum = max(body_mass_g),
    Variance = var(body_mass_g),
    Standard_Deviation = sd(body_mass_g),
    Q1 = quantile(body_mass_g, 0.25),
    Q3 = quantile(body_mass_g, 0.75),
    IQR = IQR(body_mass_g),
    Skewness = skewness(body_mass_g),
    Kurtosis = kurtosis(body_mass_g)
  )
```

body_mass_summary

A tibble: 1 × 12

N	Mean	Median	Minimum	Maximum	Variance	Standard_Deviation	Q1	Q3	IQR	Skewness	Kurtosis
<int>	<dbl>	<int>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
333	4207.057	4050	2700	6300	648372.5	805.2158	3550	4775	1225	0.4701162	2.259514

6. Species-Wise Descriptive Statistics of Body Mass

```
species_summary <- analysis_data %>%
  group_by(species) %>%
  summarise(
    N = n(),
    Mean = mean(body_mass_g),
    Median = median(body_mass_g),
    Minimum = min(body_mass_g),
    Maximum = max(body_mass_g),
    Variance = var(body_mass_g),
    Standard_Deviation = sd(body_mass_g),
    Q1 = quantile(body_mass_g, 0.25),
    Q3 = quantile(body_mass_g, 0.75),
    IQR = IQR(body_mass_g),
    Skewness = skewness(body_mass_g),
    Kurtosis = kurtosis(body_mass_g),
    .groups = "drop"
  )
```

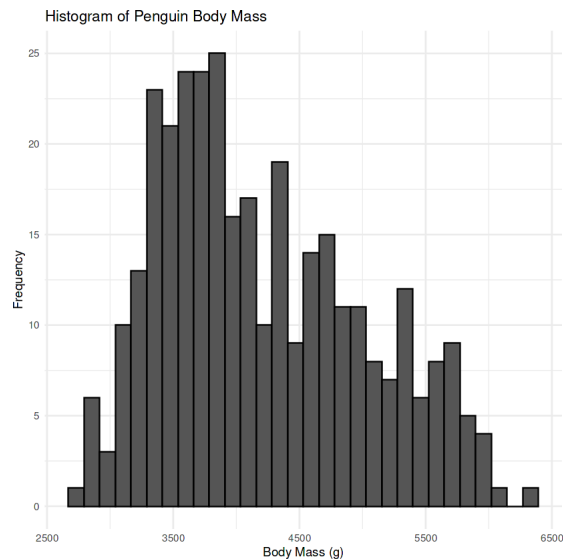
species_summary

A tibble: 3 × 13

species	N	Mean	Median	Minimum	Maximum	Variance	Standard_Deviation	Q1	Q3	IQR	Skewness	Kurtosis
<fct>	<int>	<dbl>	<dbl>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Adelie	146	3706.164	3700	2850	4775	210332.4	458.6201	3362.5	4000	637.5	0.2770077	2.411479
Chinstrap	68	3733.088	3700	2700	4800	147713.5	384.3351	3487.5	3950	462.5	0.2419413	3.463681
Gentoo	119	5092.437	5050	3950	6300	251478.3	501.4762	4700.0	5500	800.0	0.0449965	2.242850

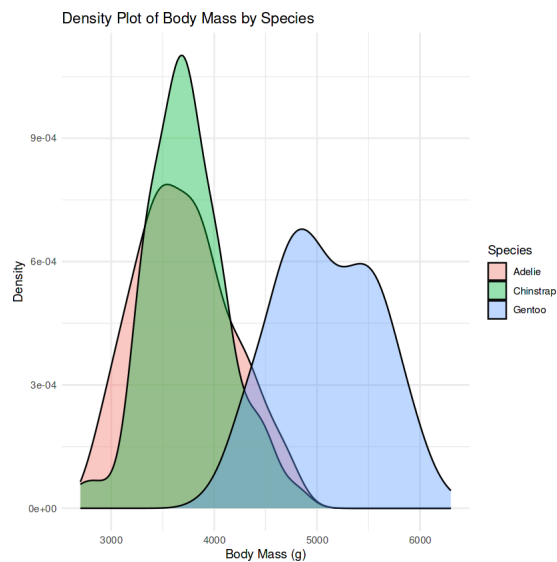
7. Histogram of Body Mass

```
ggplot(analysis_data, aes(x = body_mass_g)) +  
  geom_histogram(  
    bins = 30,  
    color = "black"  
  ) +  
  labs(  
    title = "Histogram of Penguin Body Mass",  
    x = "Body Mass (g)",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```



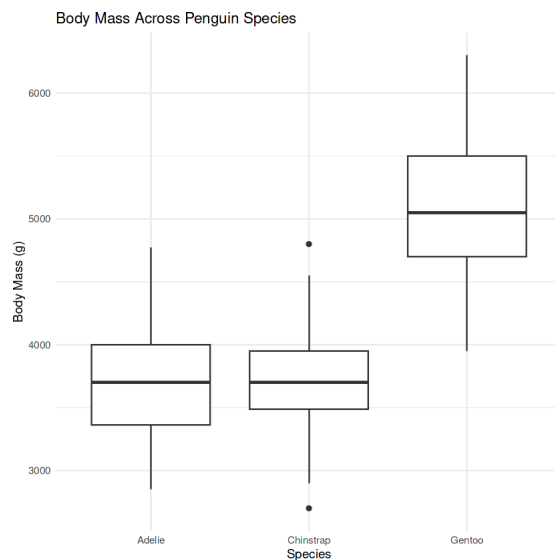
8. Density Plot of Body Mass by Species

```
ggplot(  
  analysis_data,  
  aes(x = body_mass_g, fill = species)  
) +  
  geom_density(alpha = 0.4) +  
  labs(  
    title = "Density Plot of Body Mass by Species",  
    x = "Body Mass (g)",  
    y = "Density",  
    fill = "Species"  
  ) +  
  theme_minimal()
```



9. Species-Wise Boxplot of Body Mass

```
ggplot(
  analysis_data,
  aes(x = species, y = body_mass_g)
) +
  geom_boxplot() +
  labs(
    title = "Body Mass Across Penguin Species",
    x = "Species",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



Task 2: Hypothesis Testing

Male vs Female Body Mass

Null Hypothesis (H_0)

There is no significant difference between the mean body mass of male and female penguins.

$$H_0 : \mu_{male} = \mu_{female}$$

Alternative Hypothesis (H_1)

There is a significant difference between the mean body mass of male and female penguins.

$$H_1 : \mu_{male} \neq \mu_{female}$$

The significance level is:

$$\alpha = 0.05$$

An independent two-sample Welch t-test is used because male and female penguins are independent observations.

10. Sex-Wise Descriptive Statistics

```
sex_summary <- analysis_data %>%
  group_by(sex) %>%
  summarise(
    N = n(),
    Mean = mean(body_mass_g),
    Median = median(body_mass_g),
    Standard_Deviation = sd(body_mass_g),
    Variance = var(body_mass_g),
    Q1 = quantile(body_mass_g, 0.25),
    Q3 = quantile(body_mass_g, 0.75),
    IQR = IQR(body_mass_g),
    .groups = "drop"
  )
```

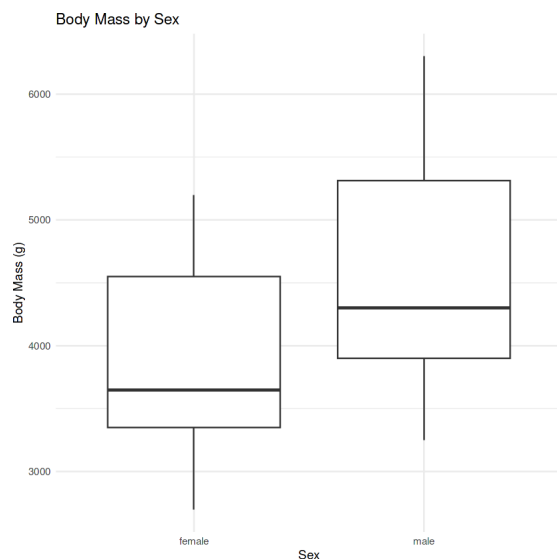
sex_summary

A tibble: 2 × 9

sex	N	Mean	Median	Standard_Deviation	Variance	Q1	Q3	IQR
<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
female	165	3862.273	3650	666.1720	443785.2	3350	4550.0	1200.0
male	168	4545.685	4300	787.6289	620359.3	3900	5312.5	1412.5

11. Sex-Wise Boxplot of Body Mass

```
ggplot(
  analysis_data,
  aes(x = sex, y = body_mass_g)
) +
  geom_boxplot() +
  labs(
    title = "Body Mass by Sex",
    x = "Sex",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



12. Shapiro-Wilk Normality Test

Hypotheses

H₀: The data are normally distributed.

H₁: The data are not normally distributed.

At $\alpha = 0.05$:

- p-value > 0.05: Fail to reject H₀
- p-value ≤ 0.05: Reject H₀

```
# Male body mass
male_mass <- analysis_data %>%
  filter(sex == "male") %>%
  pull(body_mass_g)

# Female body mass
female_mass <- analysis_data %>%
  filter(sex == "female") %>%
  pull(body_mass_g)

# Shapiro-Wilk tests
shapiro_male <- shapiro.test(male_mass)
shapiro_female <- shapiro.test(female_mass)

shapiro_male
shapiro_female
```

```
Shapiro-Wilk normality test

data:  male_mass
W = 0.92504, p-value = 1.227e-07

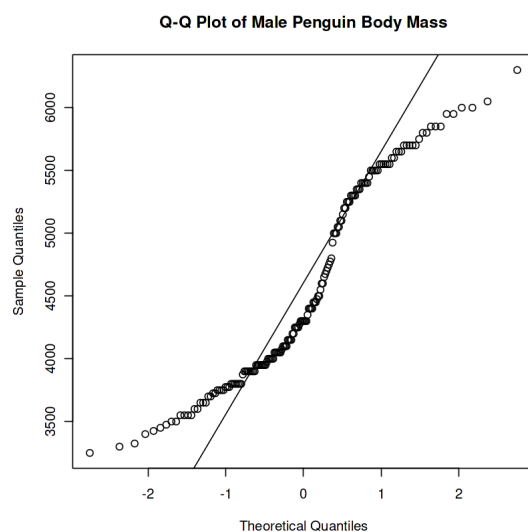
Shapiro-Wilk normality test

data:  female_mass
W = 0.91931, p-value = 6.155e-08
```

13. Q-Q Plot for Male Body Mass

```
qqnorm(
  male_mass,
  main = "Q-Q Plot of Male Penguin Body Mass"
)

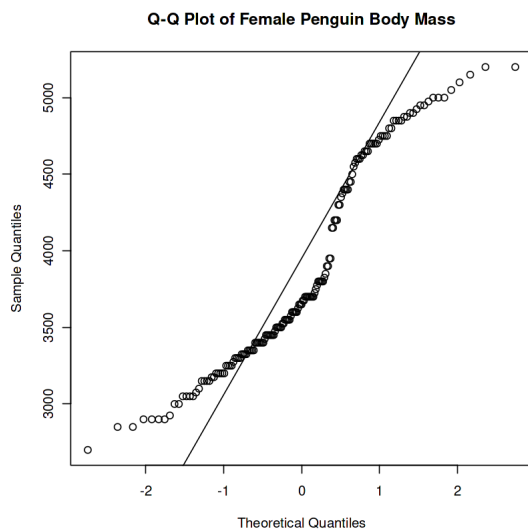
qqline(male_mass)
```



14. Q-Q Plot for Female Body Mass

```
qqnorm(
  female_mass,
  main = "Q-Q Plot of Female Penguin Body Mass"
)

qqline(female_mass)
```



15. Independent Two-Sample Welch t-Test

```
t_test <- t.test(
  body_mass_g ~ sex,
  data = analysis_data,
  var.equal = FALSE,
  conf.level = 0.95
)
```

```
t_test
```

Welch Two Sample t-test

```
data: body_mass_g by sex
t = -8.5545, df = 323.9, p-value = 4.794e-16
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -840.5783 -526.2453
sample estimates:
mean in group female    mean in group male
      3862.273           4545.685
```

16. t-Test Results

The following values are reported:

- t-statistic
- Degrees of freedom
- p-value
- 95% confidence interval
- Difference between group means

```
cat("t-statistic =", round(t_test$statistic, 4), "\n")
cat("Degrees of freedom =", round(t_test$parameter, 4), "\n")
cat("p-value =", format.pval(t_test$p.value, digits = 5), "\n")

cat(
  "95% Confidence Interval =",
  round(t_test$conf.int[1], 2),
  "to",
  round(t_test$conf.int[2], 2),
  "g\n"
)

cat(
  "Mean of first group =",
  round(t_test$estimate[1], 2),
  "g\n"
)

cat(
  "Mean of second group =",
```

```
round(t_test$estimate[2], 2),
      "g\n"
    )

t-statistic = -8.5545
Degrees of freedom = 323.8959
p-value = 4.7939e-16
95% Confidence Interval = -840.58 to -526.25 g
Mean of first group = 3862.27 g
Mean of second group = 4545.68 g
```

✓ 17. Cohen's d Effect Size

Cohen's d is used to measure the standardized difference between male and female body mass.

Interpretation

Cohen's d	Effect
0.2	Small
0.5	Medium
0.8	Large

```
# Separate male and female body mass
male_mass <- analysis_data %>%
  filter(sex == "male") %>%
  pull(body_mass_g)

female_mass <- analysis_data %>%
  filter(sex == "female") %>%
  pull(body_mass_g)

# Sample sizes
n_male <- length(male_mass)
n_female <- length(female_mass)

# Means
mean_male <- mean(male_mass)
mean_female <- mean(female_mass)

# Standard deviations
sd_male <- sd(male_mass)
sd_female <- sd(female_mass)

# Pooled standard deviation
pooled_sd <- sqrt(
  (
    (n_male - 1) * sd_male^2 +
    (n_female - 1) * sd_female^2
  ) /
  (n_male + n_female - 2)
)

# Cohen's d
cohen_d <- (mean_male - mean_female) / pooled_sd

# Display results
cat("Male mean =", round(mean_male, 2), "g\n")
cat("Female mean =", round(mean_female, 2), "g\n")
cat("Pooled SD =", round(pooled_sd, 2), "g\n")
cat("Cohen's d =", round(cohen_d, 4), "\n")
```

```
Male mean = 4545.68 g
Female mean = 3862.27 g
Pooled SD = 729.98 g
Cohen's d = 0.9362
```

```
## Interpretation of Cohen's d

cat("Cohen's d =", round(cohen_d, 4), "\n")

if (abs(cohen_d) < 0.2) {
  cat("Effect size: Negligible\n")
} else if (abs(cohen_d) < 0.5) {
  cat("Effect size: Small\n")
} else if (abs(cohen_d) < 0.8) {
  cat("Effect size: Medium\n")
} else {
  cat("Effect size: Large\n")
}
```

Cohen's $d = 0.9362$
Effect size: Large

18. Interpretation of t-Test

If the p-value is less than 0.05, the null hypothesis is rejected and there is a statistically significant difference in mean body mass between male and female penguins.

The 95% confidence interval provides the plausible range for the difference in mean body mass.

Cohen's d indicates the magnitude of the difference.

Task 3: One-Way ANOVA

The objective is to determine whether mean body mass differs among Adelie, Chinstrap, and Gentoo penguins.

Null Hypothesis (H_0)

$$\mu_{Adelie} = \mu_{Chinstrap} = \mu_{Gentoo}$$

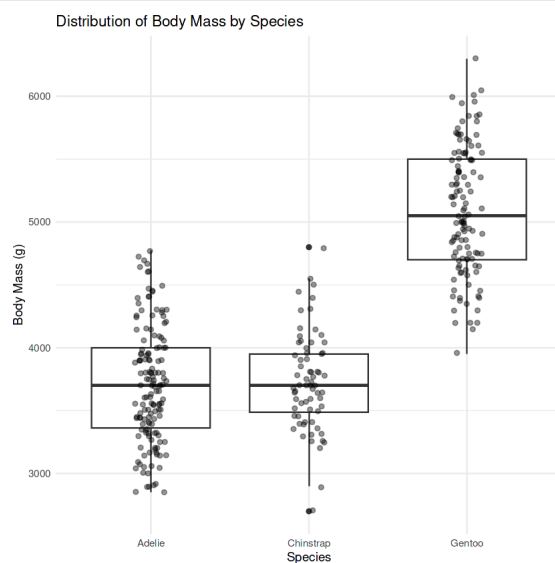
Alternative Hypothesis (H_1)

At least one species has a different mean body mass.

The significance level is $\alpha = 0.05$.

19. Distribution of Body Mass by Species

```
ggplot(
  analysis_data,
  aes(x = species, y = body_mass_g)
) +
  geom_boxplot() +
  geom_jitter(
    width = 0.1,
    alpha = 0.4
  ) +
  labs(
    title = "Distribution of Body Mass by Species",
    x = "Species",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



20. Q-Q Plots by Species

```
par(mfrow = c(1, 3))

for (sp in unique(analysis_data$species)) {
```

```

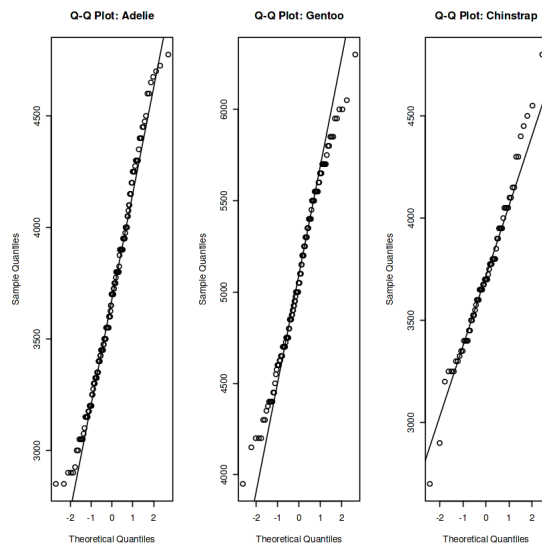
values <- analysis_data %>%
  filter(species == sp) %>%
  pull(body_mass_g)

qqnorm(
  values,
  main = paste("Q-Q Plot:", sp)
)

qqline(values)
}

par(mfrow = c(1, 1))

```



21. Shapiro-Wilk Test by Species

```

species_normality <- analysis_data %>%
  group_by(species) %>%
  shapiro_test(body_mass_g)

species_normality

```

A tibble: 3 × 4

species	variable	statistic	p
<fct>	<chr>	<dbl>	<dbl>
Adelie	body_mass_g	0.9811608	0.0423224
Chinstrap	body_mass_g	0.9844938	0.5605082
Gentoo	body_mass_g	0.9860612	0.2605247

22. Levene's Test for Homogeneity of Variance

Hypotheses

H₀: The variances of body mass are equal across species.

H₁: The variances are not equal across species.

At $\alpha = 0.05$:

- p-value > 0.05: Homogeneity assumption is supported
- p-value ≤ 0.05: Homogeneity assumption is violated

```

levene_body_mass <- leveneTest(
  body_mass_g ~ species,
  data = analysis_data
)

levene_body_mass

```

A anova: 2 × 3			
	Df	F value	Pr(>F)
	<int>	<dbl>	<dbl>
group	2	5.134899	0.006367054
	330	NA	NA

23. One-Way ANOVA

The model used is:

body_mass_g ~ species

```
one_way_anova <- aov(
  body_mass_g ~ species,
  data = analysis_data
)

summary(one_way_anova)
```

```

      Df    Sum Sq Mean Sq F value Pr(>F)
species  2 145190219 72595110   341.9 <2e-16 ***
Residuals 330  70069447  212332
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

24. ANOVA Table

The ANOVA table reports:

- Degrees of freedom
- Sum of squares
- Mean squares
- F-statistic
- p-value

```
anova_table <- summary(one_way_anova)[[1]]
```

```
anova_table
```

A anova: 2 × 5					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
species	2	145190219	72595109.6	341.8949	3.744505e-81
Residuals	330	70069447	212331.7	NA	NA

```
F_statistic <- anova_table["species", "F value"]
df_species <- anova_table["species", "Df"]
df_residual <- anova_table["Residuals", "Df"]
anova_p <- anova_table["species", "Pr(>F)"]

cat("F-statistic =", round(F_statistic, 4), "\n")
cat("Degrees of freedom =", df_species, ", ", df_residual, "\n")
cat("p-value =", format.pval(anova_p, digits = 5), "\n")
```

```
F-statistic = 341.8949
Degrees of freedom = 2 , 330
p-value = < 2.22e-16
```

25. Tukey HSD Post-Hoc Test

Tukey's HSD test is performed to identify which pairs of species differ significantly when the overall ANOVA is significant.

```
tukey_body_mass <- TukeyHSD(one_way_anova)
tukey_body_mass
```

```

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = body_mass_g ~ species, data = analysis_data)

$species
              diff            lwr            upr      p adj
Chinstrap-Adelie  26.92385 -132.3528  186.2005 0.916431
Gentoo-Adelie    1386.27259 1252.2897 1520.2554 0.000000
Gentoo-Chinstrap 1359.34874 1194.4304 1524.2671 0.000000

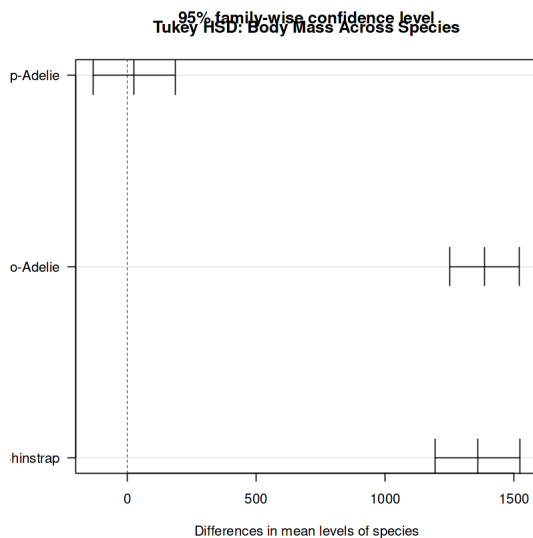
```

```

plot(
  tukey_body_mass,
  las = 1
)

title(
  main = "Tukey HSD: Body Mass Across Species"
)

```



Task 4: Non-Parametric Analysis

The Kruskal-Wallis test is used as a non-parametric alternative to one-way ANOVA.

Null Hypothesis (H_0)

The distributions of body mass are the same across all species.

Alternative Hypothesis (H_1)

At least one species has a different distribution of body mass.

26. Kruskal-Wallis Test

```

kruskal_body_mass <- kruskal.test(
  body_mass_g ~ species,
  data = analysis_data
)

```

```
kruskal_body_mass
```

Kruskal-Wallis rank sum test

```

data: body_mass_g by species
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16

```

27. ANOVA and Kruskal-Wallis Comparison

```

comparison <- data.frame(
  Test = c(
    "One-Way ANOVA",

```

```

      "Kruskal-Wallis"
    ),
    Statistic = c(
      unname(anova_table["species", "F value"]),
      unname(kruskal_body_mass$statistic)
    ),
    P_Value = c(
      anova_p,
      kruskal_body_mass$p.value
    )
  )
)

comparison

```

```

      A data.frame: 2 × 3
      Test Statistic      P_Value
      <chr>      <dbl>      <dbl>
One-Way ANOVA  341.8949  3.744505e-81
Kruskal-Wallis 212.0851  8.836877e-47

```

Interpretation

The conclusions from the One-Way ANOVA and Kruskal-Wallis test are compared.

If both tests reject the null hypothesis at $\alpha = 0.05$, the evidence that body mass differs among penguin species is consistent across parametric and non-parametric methods.

Task 5: Two-Way ANOVA

The simultaneous effects of species and sex on body mass are investigated using:

$$\text{body_mass_g} \sim \text{species} * \text{sex}$$

The model evaluates:

1. Main effect of species
2. Main effect of sex
3. Species $\times$ sex interaction

```

two_way_anova <- aov(
  body_mass_g ~ species * sex,
  data = analysis_data
)

summary(two_way_anova)

```

```

      Df    Sum Sq Mean Sq F value    Pr(>F)
species  2 145190219 72595110  758.358 < 2e-16 ***
sex       1  37090262 37090262  387.460 < 2e-16 ***
species:sex  2   1676557   838278    8.757 0.000197 ***
Residuals 327  31302628   95727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

28. Two-Way ANOVA Table

```

two_way_table <- summary(two_way_anova)[[1]]
two_way_table

```

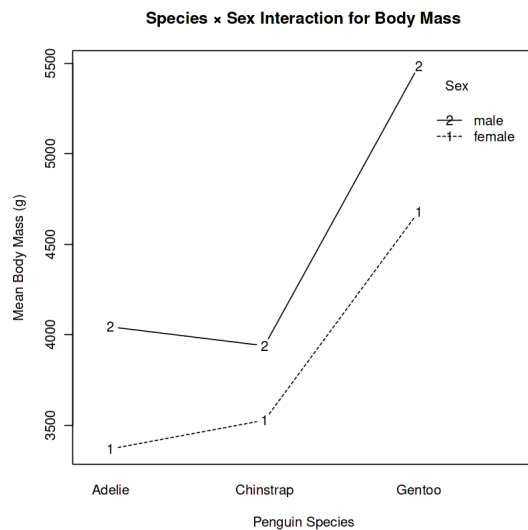
```

      A anova: 4 × 5
      Df    Sum Sq    Mean Sq    F value    Pr(>F)
      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
species      2 145190219 72595109.56  758.358072 1.540207e-123
sex           1  37090262 37090261.78  387.459976 1.902273e-57
species:sex   2   1676557   838278.37   8.756997 1.973489e-04
Residuals    327  31302628   95726.69      NA      NA

```

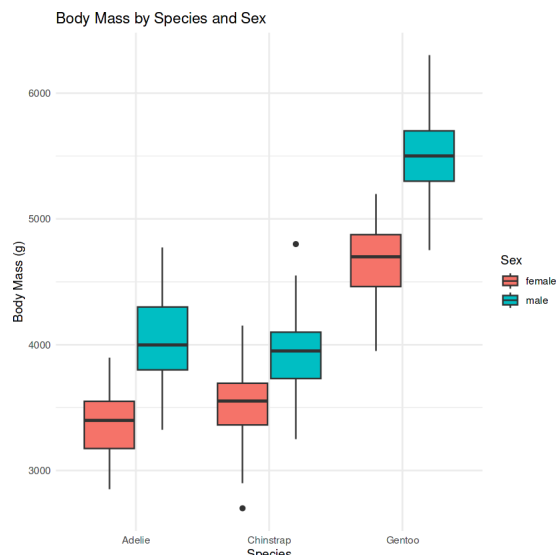
29. Species × Sex Interaction Plot

```
interaction.plot(
  x.factor = analysis_data$species,
  trace.factor = analysis_data$sex,
  response = analysis_data$body_mass_g,
  fun = mean,
  type = "b",
  xlab = "Penguin Species",
  ylab = "Mean Body Mass (g)",
  trace.label = "Sex",
  main = "Species × Sex Interaction for Body Mass"
)
```



30. Body Mass by Species and Sex

```
ggplot(
  analysis_data,
  aes(
    x = species,
    y = body_mass_g,
    fill = sex
  )
) +
  geom_boxplot(
    position = position_dodge(width = 0.8)
  ) +
  labs(
    title = "Body Mass by Species and Sex",
    x = "Species",
    y = "Body Mass (g)",
    fill = "Sex"
  ) +
  theme_minimal()
```



31. Interpretation of Two-Way ANOVA

The p-values for species, sex, and species × sex interaction are examined.

- A significant species effect indicates differences in body mass among species.
- A significant sex effect indicates differences in body mass between sexes.
- A significant species × sex interaction indicates that the effect of sex depends on species.

Task 6: Additional Analysis Using Flipper Length

The analysis is repeated using flipper length to investigate whether mean flipper length differs among the three penguin species.

Null Hypothesis (H_0)

$$\mu_{Adelie} = \mu_{Chinstrap} = \mu_{Gentoo}$$

Alternative Hypothesis (H_1)

At least one species has a different mean flipper length.

32. Descriptive Statistics of Flipper Length

```
flipper_summary <- analysis_data %>%
  group_by(species) %>%
  summarise(
    N = n(),
    Mean = mean(flipper_length_mm),
    Median = median(flipper_length_mm),
    Minimum = min(flipper_length_mm),
    Maximum = max(flipper_length_mm),
    Variance = var(flipper_length_mm),
    Standard_Deviation = sd(flipper_length_mm),
    Q1 = quantile(flipper_length_mm, 0.25),
    Q3 = quantile(flipper_length_mm, 0.75),
    IQR = IQR(flipper_length_mm),
    Skewness = skewness(flipper_length_mm),
    Kurtosis = kurtosis(flipper_length_mm),
    .groups = "drop"
  )
```

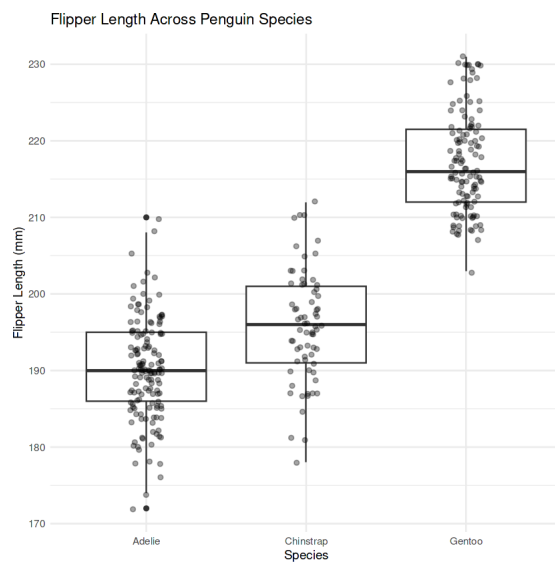
flipper_summary

A tibble: 3 × 13

species	N	Mean	Median	Minimum	Maximum	Variance	Standard_Deviation	Q1	Q3	IQR	Skewness	Kurtosis
<fct>	<int>	<dbl>	<dbl>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Adelie	146	190.1027	190	172	210	42.53420	6.521825	186	195.0	9.0	0.079578511	3.327738
Chinstrap	68	195.8235	196	178	212	50.86392	7.131894	191	201.0	10.0	-0.009262203	2.956063
Gentoo	119	217.2353	216	203	231	43.36790	6.585431	212	221.5	9.5	0.364085778	2.322009

33. Species-Wise Flipper Length Comparison

```
ggplot(
  analysis_data,
  aes(
    x = species,
    y = flipper_length_mm
  )
) +
  geom_boxplot() +
  geom_jitter(
    width = 0.1,
    alpha = 0.35
  ) +
  labs(
    title = "Flipper Length Across Penguin Species",
    x = "Species",
    y = "Flipper Length (mm)"
  ) +
  theme_minimal()
```



34. Q-Q Plots for Flipper Length

```
par(mfrow = c(1, 3))

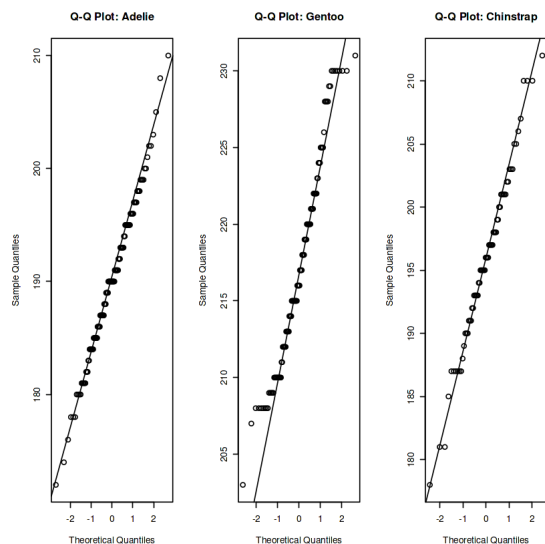
for (sp in unique(analysis_data$species)) {

  values <- analysis_data %>%
    filter(species == sp) %>%
    pull(flipper_length_mm)

  qqnorm(
    values,
    main = paste("Q-Q Plot:", sp)
  )

  qqline(values)
}

par(mfrow = c(1, 1))
```



35. Shapiro-Wilk Test for Flipper Length

```
flipper_normality <- analysis_data %>%
  group_by(species) %>%
  shapiro_test(flipper_length_mm)
```

```
flipper_normality
```

A tibble: 3 × 4

species	variable	statistic	p
<fct>	<chr>	<dbl>	<dbl>
Adelie	flipper_length_mm	0.9934049	0.742742565
Chinstrap	flipper_length_mm	0.9889115	0.810644659
Gentoo	flipper_length_mm	0.9614772	0.001759983

36. Levene's Test for Flipper Length

```
levene_flipper <- leveneTest(
  flipper_length_mm ~ species,
  data = analysis_data
)
```

```
levene_flipper
```

A anova: 2 × 3

	Df	F value	Pr(>F)
	<int>	<dbl>	<dbl>
group	2	0.4427865	0.6426253
	330	NA	NA

37. One-Way ANOVA for Flipper Length

```
flipper_anova <- aov(
  flipper_length_mm ~ species,
  data = analysis_data
)
```

```
summary(flipper_anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	50526	25263	567.4	<2e-16 ***
Residuals	330	14693	45		

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
flipper_anova_table <- summary(flipper_anova)[[1]]
```

```
flipper_anova_table
```

A anova: 2 × 5					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
species	2	50525.88	25262.94181	567.407	1.587418e-107
Residuals	330	14692.75	44.52349	NA	NA

38. Tukey HSD for Flipper Length

```
flipper_tukey <- TukeyHSD(flipper_anova)
```

```
flipper_tukey
```

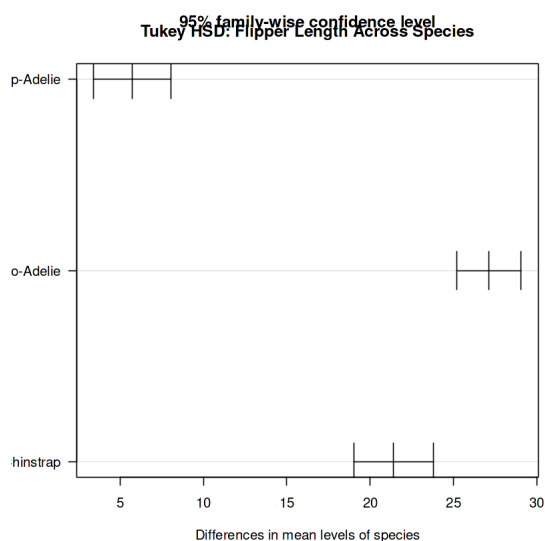
Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = flipper_length_mm ~ species, data = analysis_data)
```

```
$species
      diff      lwr      upr p adj
Chinstrap-Adelie  5.72079  3.414364  8.027215    0
Gentoo-Adelie    27.13255 25.192399 29.072709    0
Gentoo-Chinstrap 21.41176 19.023644 23.799885    0
```

```
plot(
  flipper_tukey,
  las = 1
)

title(
  main = "Tukey HSD: Flipper Length Across Species"
)
```



39. Kruskal-Wallis Test for Flipper Length

```
flipper_kruskal <- kruskal.test(
  flipper_length_mm ~ species,
  data = analysis_data
)
```

```
flipper_kruskal
```

Kruskal-Wallis rank sum test

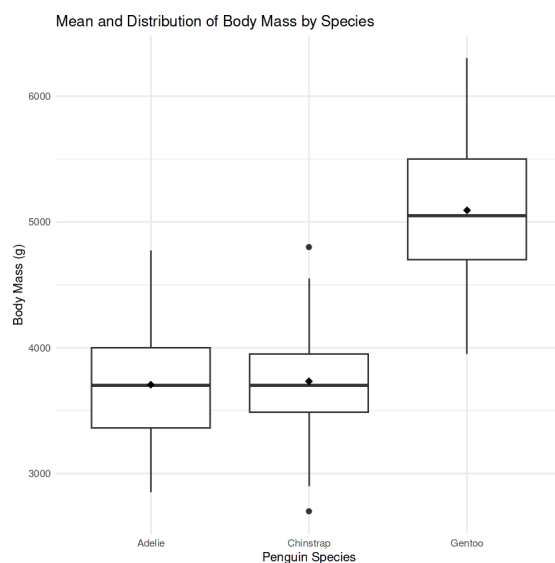
data: flipper_length_mm by species
Kruskal-Wallis chi-squared = 237.35, df = 2, p-value < 2.2e-16

Task 7: Visualization and Interpretation

The following visualizations summarize the statistical analysis:

- Histogram of body mass
- Density plot of body mass
- Species-wise body mass boxplot
- Sex-wise body mass boxplot
- Q-Q plots
- Species $\times$ sex interaction plot
- Body mass comparison by species and sex
- Species-wise flipper-length comparison
- Tukey HSD plots

```
ggplot(
  analysis_data,
  aes(
    x = species,
    y = body_mass_g
  )
) +
  geom_boxplot() +
  stat_summary(
    fun = mean,
    geom = "point",
    shape = 18,
    size = 3
  ) +
  labs(
    title = "Mean and Distribution of Body Mass by Species",
    x = "Penguin Species",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



41. Final Statistical Results

The major statistical results are summarized below.

Start coding or [generate](#) with AI.

A data.frame: 8 × 3

Analysis	Statistic	P_Value
<chr>	<chr>	<chr>
Male vs Female Body Mass	t = -8.555	4.794e-16
One-Way ANOVA: Body Mass	F = 341.895	< 2.2e-16

```
# Extract two-way ANOVA p-values
species_p <- two_way_table["species", "Pr(>F)"]
sex_p <- two_way_table["sex", "Pr(>F)"]
interaction_p <- two_way_table["species:sex", "Pr(>F)"]

# Extract flipper ANOVA values
flipper_F <- flipper_anova_table["species", "F value"]
flipper_p <- flipper_anova_table["species", "Pr(>F)"]

final_results <- data.frame(
  Analysis = c(
    "Male vs Female Body Mass",
    "One-Way ANOVA: Body Mass",
    "Kruskal-Wallis: Body Mass",
    "Two-Way ANOVA: Species",
    "Two-Way ANOVA: Sex",
    "Two-Way ANOVA: Species × Sex",
    "One-Way ANOVA: Flipper Length",
    "Kruskal-Wallis: Flipper Length"
  ),
```